

Split conformal prediction as a special case

Full Conformal	Split Conformal
All data used for training and calibration	Disjoint datasets for training and calibration
Retrains the model for each value $y \in \mathcal{Y}$	No model retraining (requires only one model fit)
Requires a symmetric score function s	Works for any (pretrained) score function s

Algorithm 3.3: Full conformal prediction

1. Input training data $(X_1, Y_1), \dots, (X_n, Y_n)$, test point X_{n+1} , target coverage level $1 - \alpha$, conformal score function s .
2. For each possible response value $y \in \mathcal{Y}$,
 - (a) Compute the score $S_i^y = s((X_i, Y_i); \mathcal{D}_{n+1}^y)$, for all $i \in [n]$.
 - (b) Compute the score for the hypothesized test point, $S_{n+1}^y = s((X_{n+1}, y); \mathcal{D}_{n+1}^y)$.
 - (c) Compute the quantile $\hat{q}^y = \text{Quantile}(S_1^y, \dots, S_n^y; (1 - \alpha)(1 + 1/n))$.
3. Return the prediction set $\mathcal{C}(X_{n+1}) = \{y \in \mathcal{Y} : S_{n+1}^y \leq \hat{q}^y\}$

Algorithm 3.6: Split conformal prediction

1. Input pretraining dataset $\mathcal{D}_{\text{pre}}$, calibration data $(X_1, Y_1), \dots, (X_n, Y_n)$, test point X_{n+1} , target coverage level $1 - \alpha$.
2. Using the pretraining dataset $\mathcal{D}_{\text{pre}}$, construct a score function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$.
3. Compute the scores on the calibration set, $S_i = s(X_i, Y_i)$ for $i \in [n]$.
4. Compute the conformal quantile $\hat{q} = \text{Quantile}(S_1, \dots, S_n; (1 - \alpha)(1 + 1/n))$.
5. Return prediction set $\mathcal{C}(X_{n+1}) = \{y \in \mathcal{Y} : s(X_{n+1}, y) \leq \hat{q}\}$

$$P(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$

- For split conformal prediction, there are two datasets at play: the calibration set D_n and the pretraining set D_{pre} .
- The vector of scores $(S_1, \dots, S_{n+1})$ depends on both datasets.
- to formally establish exchangeability of the scores, we need to consider the additional randomness coming from D_{pre} .

$H_0: (X_1, Y_1) \dots (X_n, Y_n), (X_{n+1}, Y_{n+1})$ from same dist
 Reinterpretations of conformal prediction

Definition 3.8: The conformal p-value

Given training data $(X_1, Y_1), \dots, (X_n, Y_n)$, a test feature X_{n+1} , and a score function s , the conformal p-value is defined as

Let $y = Y_{n+1}$

$$p^y = \frac{1 + \sum_{i=1}^n \mathbb{1}\{S_i^y \geq S_{n+1}^y\}}{n+1}$$

$S_1^y \dots S_n^y, S_{n+1}^y$

for each $y \in \mathcal{Y}$, where as before,

$$S_i^y = s((X_i, Y_i); \mathcal{D}_{n+1}^y), i \in [n], \quad S_{n+1}^y = s((X_{n+1}, y); \mathcal{D}_{n+1}^y).$$

$$\underbrace{\sum_{i=1}^{n+1} \mathbb{1}\{S_i^y \geq S_{n+1}^y\}}_{n+1}$$

This p-value is asking whether the hypothesized test point (X_{n+1}, y) appears to follow the same distribution as the training data $(X_1, Y_1), \dots, (X_n, Y_n)$.

we can interpret p^y as a p-value testing the hypothesis of exchangeability of the data points $(X_1, Y_1), \dots, (X_n, Y_n), (X_{n+1}, y)$.

repeat this reasoning for every possible value $y \in \mathcal{Y}$, and collect all the plausible values into a prediction set.

Conformal prediction is exactly equivalent to this procedure:

Proposition 3.9: Full conformal prediction via the conformal p-value

The full conformal prediction set $\mathcal{C}(X_{n+1})$ defined in (3.2) satisfies

$$\mathcal{C}(X_{n+1}) = \{y \in \mathcal{Y} : p^y > \alpha\},$$

where p^y is the conformal p-value (Definition 3.8).

P is valid ? $\underline{P(p \leq \alpha; H_0) \leq \alpha}$

$$y \notin \mathcal{C}(X_{n+1}) = \{y : S_{n+1}^y \leq \text{Quantile}(S_1^y, \dots, S_n^y, (1-\alpha)(1+1/n))\}$$

Proof of Proposition 3.9

First, by definition of the full conformal prediction set,

$$y \notin \mathcal{C}(X_{n+1}) \iff S_{n+1}^y > \text{Quantile}(S_1^y, \dots, S_n^y, (1-\alpha)(1+1/n)).$$

Next, by definition of the quantile of a finite list, for any $\tau \in [0, 1]$ and any $t \in \mathbb{R}$ it holds that

$$\text{Quantile}(S_1^y, \dots, S_n^y; \tau) < t \iff \sum_{i=1}^n \mathbb{1}\{S_i^y < t\} \geq n\tau.$$

$$\text{Quantile}(S_1, \dots, S_n; \tau) = \inf\{v : \hat{F}(v) \geq \tau\} \quad \text{if } Q < t \Rightarrow \hat{F}(t) \geq \tau$$

$$\text{if } Q < t \Rightarrow \hat{F}(t) \geq \tau$$

Choosing $\tau = (1-\alpha)(1+1/n)$, then, we have

$$t = S_{n+1}^y$$

$$y \notin \mathcal{C}(X_{n+1}) \iff \sum_{i=1}^n \mathbb{1}\{S_i^y < S_{n+1}^y\} \geq (1-\alpha)(n+1).$$

$$\iff \sum_{i=1}^n \mathbb{1}\{S_i^y < t\} \geq n\tau$$

Finally, we return to the p-value: we calculate

$$p^y = \frac{1 + \sum_{i=1}^n \mathbb{1}\{S_i^y \geq S_{n+1}^y\}}{n+1} = 1 - \frac{\sum_{i=1}^n \mathbb{1}\{S_i^y < S_{n+1}^y\}}{n+1}, \leq \alpha$$

i.e., $p^y \leq \alpha$ if and only if $y \notin \mathcal{C}(X_{n+1})$, as desired.

$$\mathbb{1}\{S_i^y \geq S_{n+1}^y\} = 1 - \mathbb{1}\{S_i^y < S_{n+1}^y\}$$

Proof of Theorem 3.2 via permutation tests

For convenience, throughout this proof we will write $Z_i = (X_i, Y_i)$ for the i th data point, and will write $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$.

First we recall the permutation test p-value, developed in Section 2.2: given any test function $T : \mathcal{Z}^{n+1} \rightarrow \mathbb{R}$, we define

$$p_{\text{perm}} = \frac{\sum_{\sigma \in \mathcal{S}_{n+1}} \mathbb{1} \{T(Z_{\sigma(1)}, \dots, Z_{\sigma(n+1)}) \geq T(Z_1, \dots, Z_{n+1})\}}{(n+1)!},$$

where, as before, $\mathcal{S}_{n+1}$ denotes the set of permutations on $[n+1] = \{1, \dots, n+1\}$. (Note that here, we are working with a vector of length $n+1$, rather than n as in the original definition of this p-value in (2.3).) If the combined training and test dataset $(Z_1, \dots, Z_{n+1})$ is indeed exchangeable, then we have $\mathbb{P}(p_{\text{perm}} \leq \alpha) \leq \alpha$, as established in Theorem 2.4.

To complete the proof, then, we will verify that

$$Y_{n+1} \in \mathcal{C}(X_{n+1}) \iff p_{\text{perm}} > \alpha,$$

when the test function T is chosen appropriately. In particular, in light of Proposition 3.9, it suffices to show that

$$p_{\text{perm}} = p^{Y_{n+1}}.$$

We will choose the test function T as

$$T(z_1, \dots, z_{n+1}) = s(z_{n+1}; (z_1, \dots, z_n)), = |y - f(x_{n+1})|$$

i.e., the score for the last data point z_{n+1} when trained on a dataset $(z_1, \dots, z_n)$. In particular, we can calculate the value of this test function on the permuted dataset $(Z_{\sigma(1)}, \dots, Z_{\sigma(n+1)})$ as

$$A = T(Z_{\sigma(1)}, \dots, Z_{\sigma(n+1)}) = s(Z_{\sigma(n+1)}; (\mathcal{D}_{n+1})_{\sigma}) = s(Z_{\sigma(n+1)}; \mathcal{D}_{n+1}) = S_{\sigma(n+1)},$$

where the first step holds since $(\mathcal{D}_{n+1})_{\sigma} = (Z_{\sigma(1)}, \dots, Z_{\sigma(n+1)})$ by definition, and the second step holds since s is assumed to be symmetric. Therefore,

$$p_{\text{perm}} = \frac{\sum_{\sigma \in \mathcal{S}_{n+1}} \mathbb{1} \{S_{\sigma(n+1)} \geq S_{n+1}\}}{(n+1)!} = \sum_{i=1}^{n+1} n! \cdot \frac{\mathbb{1} \{S_i \geq S_{n+1}\}}{(n+1)!},$$

$$\sum_{i=1}^{n+1} \sum_{\sigma(n+1)=i} \frac{\mathbb{1} \{S_i \geq S_{n+1}\}}{(n+1)!}$$

where the last step holds since, for each $i \in [n+1]$, there are exactly $n!$ many permutations $\sigma \in \mathcal{S}_{n+1}$ for which $\sigma(n+1) = i$. After simplifying, we then have

$$p_{\text{perm}} = \frac{\sum_{i=1}^{n+1} \mathbb{1} \{S_i \geq S_{n+1}\}}{n+1} = \frac{1 + \sum_{i=1}^n \mathbb{1} \{S_i \geq S_{n+1}\}}{n+1} = p^{Y_{n+1}},$$

with the last step holding since $S_i = S_i^{Y_{n+1}}$ for all i by definition. This establishes (3.10) as desired.

$$\mathbb{P}(p^{Y_{n+1}} \leq \alpha) \leq \alpha$$

$Z_1, \dots, Z_{n+1}$ are exchangeable (2) $Z_{n+1} | \hat{P}_{n+1} \sim \hat{P}_{n+1}$
 $= \frac{1}{n+1} \sum_{i=1}^{n+1} \delta_{Z_i}$

Another interpretation of conformal prediction is given by treating the dataset as an unordered collection of data points, and then using exchangeability to argue that the test point (X_{n+1}, Y_{n+1}) is equally likely to be any one of these points.

equally likely to be any one of these points. In particular, if each data point $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ is distinct, and if we observe the unordered set of $n+1$ values (i.e., we observe the unordered set of data points $\{(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})\}$, but do not know which data value corresponds to which index $i \in [n+1]$), then the test point (X_{n+1}, Y_{n+1}) is equally likely to be any one of these $n+1$ values.

To formalize this intuition, given the dataset $\mathcal{D}_{n+1} = ((X_1, Y_1), \dots, (X_{n+1}, Y_{n+1}))$ containing the training data and test point, recalling our notation from Chapter 2 we define its empirical distribution as

$$\hat{P}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} \delta_{(X_i, Y_i)}$$

$H_0: Y = y$ $H_0: Y \geq 50$

Proof of Theorem 3.2 via the empirical distribution of the data

First, since the score function s is assumed to be symmetric, note that $s(\cdot; \mathcal{D}_{n+1})$ depends on the dataset $\mathcal{D}_{n+1}$ only via the empirical distribution $\hat{P}_{n+1}$. To emphasize this point, we will use the notation $s(\cdot; \hat{P}_{n+1})$ in place of $s(\cdot; \mathcal{D}_{n+1})$ throughout the remainder of this proof. We will also write $Z_i = (X_i, Y_i)$ for the i th data point, and $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$.

Let $h: \mathcal{Z} \rightarrow \mathbb{R}$ be any function. Conditioning on $\hat{P}_{n+1}$, by Proposition 2.2, the test point Z_{n+1} has distribution $\hat{P}_{n+1}$. This implies that, again conditioning on $\hat{P}_{n+1}$, the random variable $h(Z_{n+1})$ has distribution $\frac{1}{n+1} \sum_{i=1}^{n+1} \delta_{h(Z_i)}$. In particular,

$$\mathbb{P}(h(Z_{n+1}) \leq \text{Quantile}((h(Z_i))_{i \in [n+1]}; 1 - \alpha) | \hat{P}_{n+1}) \geq 1 - \alpha,$$

by definition of the quantile. Next, observe that since this calculation is carried out conditionally on $\hat{P}_{n+1}$, the same argument holds for a function h that depends on the empirical distribution $\hat{P}_{n+1}$ as well as on a data point $z \in \mathcal{Z}$ —in particular, we can take $h(z) = s(z; \hat{P}_{n+1})$, to obtain

$$\mathbb{P}(s(Z_{n+1}; \hat{P}_{n+1}) \leq \text{Quantile}((s(Z_i; \hat{P}_{n+1}))_{i \in [n+1]}; 1 - \alpha) | \hat{P}_{n+1}) \geq 1 - \alpha.$$

$$Z_{n+1} | \hat{P}_{n+1} \sim \hat{P}_{n+1}$$

$$P(Z_{n+1} \leq \text{Quantile}((Z_i)_{i \in [n+1]}; 1 - \alpha) | \hat{P}_{n+1}) \geq 1 - \alpha$$

$$Z_{n+1} | Z_1, \dots, Z_n, Z_{n+1} \sim \hat{P}_{n+1}$$

$$\Leftrightarrow Z_1, \dots, Z_{n+1} \text{ are exchangeable}$$

$$S(z; D_{n+1}) = |y - f(x)| = s(z; \hat{P}_{n+1})$$

Since $S_i = s(Z_i; \mathcal{D}_{n+1}) = s(Z_i; \hat{P}_{n+1})$ for each $i \in [n+1]$, by definition, we then have

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \hat{P}_{n+1}) = \mathbb{P}\left(S_{n+1} \leq \text{Quantile}(S_1, \dots, S_{n+1}; 1 - \alpha) \mid \hat{P}_{n+1}\right) \geq 1 - \alpha,$$

where the first step applies (3.5). Marginalizing over $\hat{P}_{n+1}$, we have proved the claim.

$$\mathbb{P}(S_{n+1} \leq Q) \geq 1 - \alpha$$

$$= \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$